

Math Thermo
class # 03
01/27/2026

Differential Notation

In the physics literature it is common to write

$$(3.1) \quad d\tilde{U}^\alpha = T_v^\alpha dS^\alpha - P_v^\alpha dV^\alpha + \sum_{i=1}^r \mu_{v,i}^\alpha dN_i^\alpha$$

and

$$(3.2) \quad d\tilde{S}^\alpha = \frac{1}{T_s^\alpha} dU^\alpha + \frac{P_s^\alpha}{T_s^\alpha} dV^\alpha - \sum_{i=1}^r \frac{\mu_{s,i}^\alpha}{T_s^\alpha} dN_i^\alpha.$$

These are referred to as the combined 1st & 2nd Laws. Here,

(3.1) means, precisely, $\tilde{U}^\alpha = \tilde{U}^\alpha(S, V, \vec{N})$ and

$$\frac{\partial \tilde{U}^\alpha}{\partial S^\alpha} = T_v^\alpha, \quad \frac{\partial \tilde{U}^\alpha}{\partial V^\alpha} = -P_v^\alpha, \quad \frac{\partial \tilde{U}^\alpha}{\partial N_j^\alpha} = \mu_{v,i}^\alpha,$$

and (3.2) simply means $\tilde{S}^\alpha = \tilde{S}^\alpha(U, V, \vec{N}^\alpha)$ with

$$\frac{\partial \tilde{S}^\alpha}{\partial U^\alpha} = \frac{1}{T_s^\alpha}, \quad \frac{\partial \tilde{S}^\alpha}{\partial V^\alpha} = \frac{P_s^\alpha}{T_s^\alpha}, \quad \frac{\partial \tilde{S}^\alpha}{\partial N_j^\alpha} = -\frac{\mu_{s,i}^\alpha}{T_s^\alpha}.$$

We will ascribe real meaning to these relations, once we define process paths, using the chain rule in $\mathbb{R}^n$.

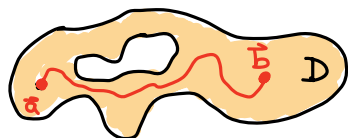
Path (Contour) Integrals

Defn (3.1): Suppose that $D \subseteq \mathbb{R}^n$ is open.

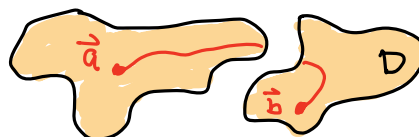
A function $\vec{\gamma}: [a, b] \rightarrow D$ is called a path (or contour) iff $\vec{\gamma}$ is continuous and

piecewise smooth. D is called path-connected iff for every two distinct points $\vec{a}, \vec{b} \in D$ there is path $\vec{\gamma}: [a, b] \rightarrow D$ such that

$$\vec{\gamma}(a) = \vec{a} \quad \vec{\gamma}(b) = \vec{b}.$$



Path-Connected

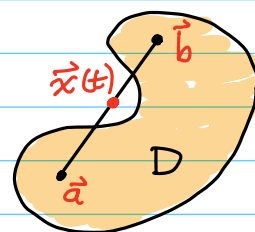


Not Path Connected

D is called Simply-connected iff it is (1) path connected and (2) every closed path can be continuously deformed to a point, i.e., there are "no holes." D is called convex iff for every pair $\vec{a}, \vec{b} \in D$, the point

$$\vec{x}(t) = \vec{a}(1-t) + \vec{b}t \in D,$$

for all $t \in [0, 1]$.



Definition (3.2): Let $\vec{F}: D \rightarrow \mathbb{R}^n$ be a C^0 function, i.e., $\vec{F} \in C^0(D; \mathbb{R}^n)$. Let $\vec{\gamma}: [a, b] \rightarrow D$ be a path in D , which is assumed to be simply connected. Then the path integral $\int_{\vec{\gamma}} \vec{F}(\vec{x}) \cdot d\vec{x}$ is defined via

$$(3.3) \quad \int_{\vec{\gamma}} \vec{F}(\vec{x}) \cdot d\vec{x} := \int_a^b \vec{F}(\vec{\gamma}(\tau)) \cdot \vec{\gamma}'(\tau) d\tau.$$

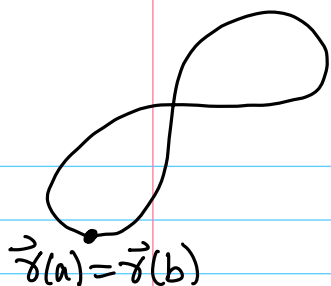
We will also use the notation

$$\int_{\vec{\gamma}} \vec{F}(\vec{x}) \cdot d\vec{x} = \int_{\vec{\gamma}} F_1(\vec{x}) dx_1 + \dots + F_n(\vec{x}) dx_n.$$

Defn (3.3): Let $D \subseteq \mathbb{R}^n$ be a simply connected open set. A path $\vec{\gamma}: [a, b] \rightarrow D$ is called closed iff

$$\vec{\gamma}(a) = \vec{\gamma}(b)$$

A closed path is called simple iff it does not intersect itself except at $\tau = a$ and $\tau = b$, i.e., for every $c \in (a, b)$,



$$\vec{\gamma}(c) \neq \vec{\gamma}(u), \quad u \in [a, c) \cup (c, b].$$

Theorem (3.4): Let $D \subseteq \mathbb{R}^n$ be an open, simply-connected set. Assume $\vec{\gamma}: [a, b] \rightarrow D$ is a path. Suppose $\eta: [c, d] \rightarrow [a, b]$ is a monotonically increasing, C^1 function with $\eta(c) = a$, $\eta(d) = b$. Define

$$\vec{\chi}(t) := \vec{\gamma}(\eta(t)), \quad \forall t \in [c, d].$$

Then

$$\int_{\vec{\gamma}} \vec{F}(\vec{x}) \cdot d\vec{x} = \int_{\vec{\chi}} \vec{F}(\vec{x}) \cdot d\vec{x}.$$

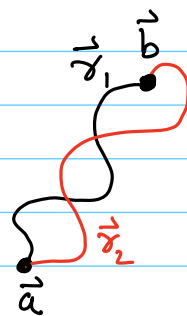
Proof: An easy exercise using change-of-variables. |||

This result guarantees that the path integrals are parameterization independent.

Dfn (3.5): Let $D \subseteq \mathbb{R}^n$ be an open, simply-connected set. Suppose that

$$\int_{\vec{\gamma}_1} \vec{F}(\vec{x}) \cdot d\vec{x} = \int_{\vec{\gamma}_2} \vec{F}(\vec{x}) \cdot d\vec{x}$$

for any two paths $\vec{\gamma}_1: [a, b] \rightarrow D$, $\vec{\gamma}_2: [c, d] \rightarrow D$ with



$$\vec{a} = \vec{\gamma}_1(a) = \vec{\gamma}_2(a) \quad \text{and} \quad \vec{\gamma}_1(b) = \vec{\gamma}_2(b) = \vec{b}.$$

Then we say that the integral is path independent. Note that we are not assuming that

$$\text{Range}(\vec{\gamma}_1) = \text{Range}(\vec{\gamma}_2).$$

Defn (3.6): let D be an open set and $\vec{F} \in C^0(D; \mathbb{R}^n)$. We say that $\vec{F}$ is conservative iff there is a function $f \in C^1(D; \mathbb{R})$ such that

$$\vec{F}(\vec{x}) = \nabla f(\vec{x}), \quad \forall \vec{x} \in D.$$

Theorem (3.7): let D be an open simply connected set in $\mathbb{R}^n$. If $\vec{F} \in C^0(D; \mathbb{R}^n)$ is conservative, then the integral

$$\int_{\vec{\gamma}} \vec{F}(\vec{x}) \cdot d\vec{x}$$

is path-independent.

Proof: let $\vec{\gamma}_1: [a_1, b_1] \rightarrow D$ and $\vec{\gamma}_2: [a_2, b_2] \rightarrow D$ be paths in D with the same end points, i.e.,

$$\vec{a} := \vec{\gamma}_1(a_1) = \vec{\gamma}_2(a_2), \quad \vec{\gamma}_1(b_1) = \vec{\gamma}_2(b_2) =: \vec{b}$$

By the chain rule, for $i=1,2$,

$$\begin{aligned} \frac{d}{dt} f(\vec{\gamma}_i(t)) &= \nabla f(\vec{\gamma}_i(t)) \cdot \vec{\gamma}_i'(t) \\ &= \vec{F}(\vec{\gamma}_i(t)) \cdot \vec{\gamma}_i'(t) \end{aligned}$$

Thus,

$$\int_{\vec{\gamma}_1} \vec{F}(\vec{x}) \cdot d\vec{x} = \int_{a_1}^{b_1} \vec{F}(\vec{\gamma}_1(t)) \cdot \vec{\gamma}_1'(t) dt$$

$$= \int_{a_i}^{b_i} \frac{d}{d\tau} [f(\vec{x}_i(\tau))] d\tau$$

$$\stackrel{\text{FTC}}{=} f(\vec{b}) - f(\vec{a}),$$

for $i=1, 2$.

We have the following well-known results.

Theorem (3.8): Let $D \subseteq \mathbb{R}^n$ be a simply-connected set and suppose that $\vec{F} \in C^0(D; \mathbb{R}^n)$. The following are equivalent:

- 1) $\vec{F}$ is conservative ($\vec{F} = \nabla f \exists f \in C^1(D, \mathbb{R})$)
- 2) $\int_{\gamma} \vec{F}(\vec{x}) d\vec{x}$ is path independent
- 3) $\oint_{\gamma} \vec{F}(\vec{x}) d\vec{x} = 0$ for any closed path.

Recall, we wrote, as a short hand

$$d\tilde{U} = T_v dS - P_v dV + \sum_{i=1}^r \mu_{v,i} dN_i$$

This has the form

$$F_1 dx_1 + F_2 dx_2 + \dots + F_n dx_n = \vec{F} \cdot d\vec{x}$$

where

$$F_1 = T_v, F_2 = -P_v, \dots$$

We say that $\vec{F} \cdot d\vec{x}$ is an exact differential if $\vec{F}$ is conservative.

Clearly
$$d\tilde{U} = \nabla \tilde{U} \cdot d\vec{\sigma} \quad (\vec{\sigma} \in \Sigma_s)$$

is an exact differential, because $\nabla \tilde{U}$ is conservative, trivially. Thus, the integral

$$\int_{\vec{\gamma}} d\tilde{U} = \int_{\vec{\gamma}} \nabla \tilde{U} \cdot d\vec{\sigma}$$

is path independent. If $\vec{\gamma}: [a, b] \rightarrow \Sigma_s$ is the path in question, with

$$\vec{\gamma}(a) = \vec{\sigma}_a, \quad \vec{\gamma}(b) = \vec{\sigma}_b.$$

then

$$\int_{\vec{\gamma}} d\tilde{U} = \int_{\vec{\gamma}} \nabla \tilde{U} \cdot d\vec{\sigma} = \tilde{U}(\vec{\sigma}_b) - \tilde{U}(\vec{\sigma}_a).$$

We can use any path we want in state space Σ_s

The same is true for

$$d\tilde{S} = \frac{1}{T_s} dU + \frac{P_s}{T_s} dV - \sum_{i=1}^r \frac{\mu_{s,i}}{T_s} dN_i,$$

that is

$$d\tilde{S} = \nabla \tilde{S} \cdot d\vec{\sigma} \quad (\vec{\sigma} \in \Sigma_v)$$

is an exact differential.

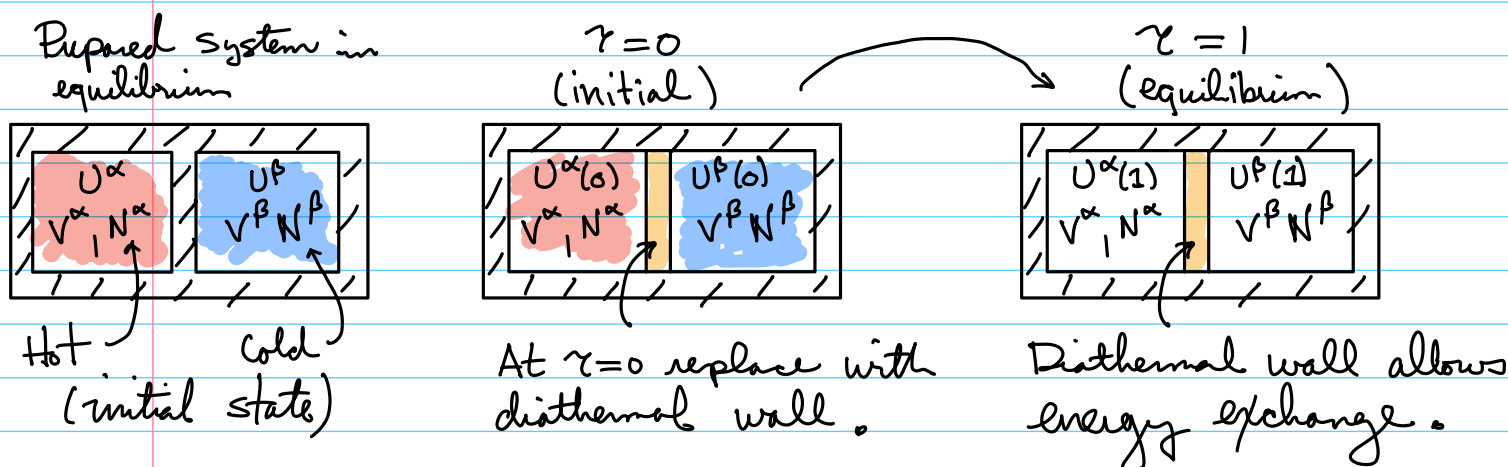
Heat Flow

Let us examine the approach to thermal equilibrium using a diathermal wall between two otherwise isolated systems α and β . Suppose the substance is unary, $r=1$. It must be that the variables

$$V^\alpha, V^\beta, N^\alpha, N^\beta$$

are fixed. But energy can be exchanged in the process.

Suppose the variable τ parameterizes the process path.



$$\begin{aligned}
 & \text{initial state} \\
 S(0) &= \tilde{S}^\alpha(U^\alpha(0), V^\alpha, N^\alpha) + \tilde{S}^\beta(U^\beta(0), V^\beta, N^\beta) \\
 & \leq \underbrace{\tilde{S}^\alpha(U^\alpha(1), V^\alpha, N^\alpha) + \tilde{S}^\beta(U^\beta(1), V^\beta, N^\beta)}_{\text{equilibrium state}} = S(1)
 \end{aligned}$$

Process Paths

$$\vec{\gamma}^\alpha(r) = \begin{bmatrix} U^\alpha(r) \\ V^\alpha \\ N^\alpha \end{bmatrix}, \quad \vec{\gamma}^\beta(r) = \begin{bmatrix} U^\beta(r) \\ V^\beta \\ N^\beta \end{bmatrix}.$$

We require that

$$U^\alpha(r) + U^\beta(r) = U_0 > 0.$$

The entropy is at a max at equilibrium.

Suppose that, as indicated by the figure above,

$$(\text{hot}) \quad T_s^\alpha(U^\alpha(0), V^\alpha, N^\alpha) > T_s^\beta(U^\beta(0), V^\beta, N^\beta), \quad (\text{cold})$$

and we know that when equilibrium is attained

$$T_s^\alpha(U^\alpha(1), V^\alpha, N^\alpha) = T_s^\beta(U^\beta(1), V^\beta, N^\beta).$$

We will show that

$$U^\alpha(0) > U^\alpha(1) \quad (\text{energy is lost in } \alpha)$$

and

$$U^\beta(0) < U^\beta(1) \quad (\text{energy is gained in } \beta)$$

To see this compute

$$\begin{aligned} \frac{dS}{dr} &= \frac{\partial \tilde{S}^\alpha}{\partial U^\alpha} \frac{\partial U^\alpha}{\partial r} + \frac{\partial \tilde{S}^\beta}{\partial U^\beta} \frac{\partial U^\beta}{\partial r} \\ &= \frac{1}{T_s^\alpha} \frac{\partial U^\alpha}{\partial r} + \frac{1}{T_s^\beta} \frac{\partial U^\beta}{\partial r} \\ &= \left(\frac{1}{T_s^\alpha(r)} - \frac{1}{T_s^\beta(r)} \right) \frac{\partial U^\alpha}{\partial r} \end{aligned}$$

(3.4)

where

$$T_s^q(r) := T_s^q(U^q(r), V^q, N^q), \quad q = \alpha, \beta.$$

We know that

$$S(0) \leq S(1).$$

We also know that

$$T_S^\alpha(0) > T_S^\beta(0)$$

and

$$T_S^\alpha(1) = T_S^\beta(0).$$

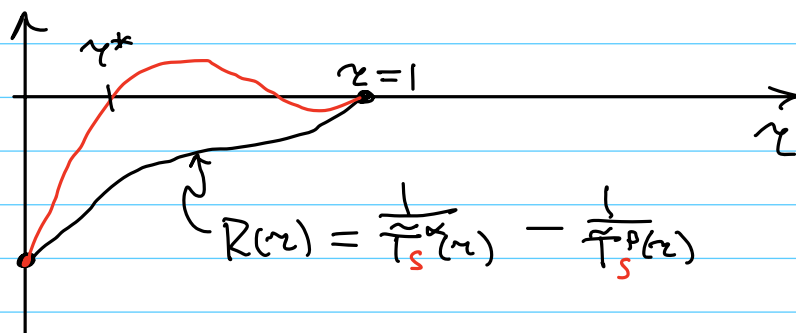
Set

$$R(r) := \frac{1}{\tilde{T}_S^\alpha(r)} - \frac{1}{\tilde{T}_S^\beta(r)} \quad 0 \leq r \leq 1$$

It follows that

$$R(0) < 0 \quad \text{and} \quad R(1) = 0.$$

Thus,



The red curve is not possible; if it were equilibrium would be reached earlier, at $r=r^*$.
Thus,

$$R(r) < 0 \quad \forall r \in [0, 1).$$

Integrating equation (3.4), we get

$$(3.5) \quad 0 \leq S(1) - S(0) = \int_0^1 R(r) \frac{dU^\alpha}{dr} dr$$

For $U^\alpha(r)$ we have two options

$$\text{Case (1)} \quad U^\alpha(0) > U^\alpha(1)$$

or

$$\text{Case (2)} \quad U^\alpha(0) \leq U^\alpha(1).$$

path indep. $\Rightarrow$

We are free to pick a process path however we want. Let us take a simple linear path

$$\frac{dU^\alpha}{dr} = \frac{U^\alpha(1) - U^\alpha(0)}{1} =: C^\alpha$$

Then

$$\text{Case (1)} \Rightarrow C^\alpha < 0$$

$$\text{Case (2)} \Rightarrow C^\alpha \geq 0$$

The integral in (3.5) is

$$0 \leq S(1) - S(2) = C^\alpha \underbrace{\int_0^1 R(r) dr}_{< 0}.$$

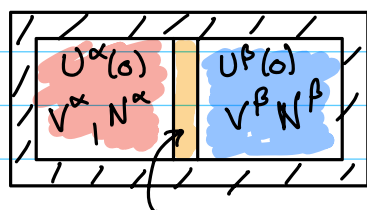
Therefore, the only possible choice is Case (1).

Thus

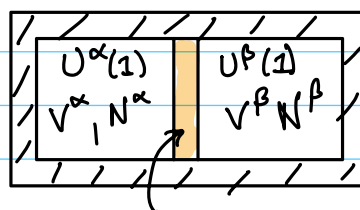
$$U^\alpha(0) > U^\alpha(1) \quad (U^\beta(0) < U^\beta(1)).$$

This is consistent with our intuition about temperature. If $T_s^\alpha(0) > T_s^\beta(0)$ heat energy flows from subsystem α to subsystem β :

$r=0$
(initial)



Diathermal wall.



Diathermal wall.

$r=1$
(equilibrium)

Heat Transfer Principle: Heat energy always flows from hotter to colder systems. This is equivalent to the second law of thermodynamics.